

Capstone Project Proposal

- **Group Details and Project Title**

Student Details (please add details for all members)

SN	Name	NAPS ID	Email	Phone
1	To be allocated	N/A	N/A	N/A

To be completed after student group allocation.

Title of Project: Design and Development of a Cybersecurity Awareness Training and Quiz System

Type of Project: Tick one:

- ☐ Research-based
- ☐ Software development
- ☐ AI/Data science
- ☒ **Cybersecurity**
- ☐ Industry consultancy
- ☐ Other

- **Course:** BIT/MIT

- **Project Details**

1. Client Overview

The proposed internal client is the NAPS School of IT. The project is intended to provide a simple cybersecurity awareness training and quiz platform for students, staff, or general users. The system will help users learn basic cybersecurity practices through short learning modules, sample scenarios, quizzes, scores, and feedback. The project is designed to be practical, safe, and easy for BIT/MIT students to implement and demonstrate using sample content.

Organisation Name: NAPS - School of IT

Contact Person and Role: Rajan Kadel - Capstone Project Coordinator / Academic Staff

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2. Client Needs and Problem Context

Cybersecurity awareness is important because many security incidents begin with simple user mistakes, such as weak passwords, unsafe links, suspicious attachments, or poor handling of personal information. The client needs a simple educational prototype that can present basic cybersecurity topics, test user understanding, and show progress or quiz results. The project does not require advanced security tools, live phishing campaigns, or complex machine learning. It should focus on awareness, learning, quiz management, reporting, and a clear final demonstration.

3. Project Purpose and Objectives

The purpose of this project is to design and develop a simple web-based cybersecurity awareness training and quiz system that helps users learn basic online safety concepts and check their understanding through short quizzes.

Objectives:

- Design a simple database for users, training topics, quiz questions, answers, scores, and completion status.
- Develop user screens for viewing training modules, attempting quizzes, and receiving feedback.
- Develop administrator screens for adding or editing topics, questions, and quiz options.
- Create simple reports showing user scores, completed modules, and areas where users need improvement.
- Test the prototype using sample users and sample cybersecurity awareness content.

4. High-Level Requirements

The system should be simple, safe, and easy to demonstrate. The following high-level requirements are recommended:

- Basic login or demo user access for learner and administrator roles.
- Training module pages covering simple topics such as password safety, phishing awareness, safe browsing, multi-factor authentication, and data privacy.
- Multiple-choice quiz functionality for each training module.
- Automatic calculation of quiz scores and basic feedback after completion.
- Administrator function to add, edit, or delete training topics and quiz questions.
- User progress page showing completed modules, scores, and quiz attempts.
- Simple dashboard or report page showing total users, average score, pass/fail status, and most difficult topics.
- Optional printable completion certificate or completion summary for demonstration purposes.

5. Key Stakeholders

The main stakeholders for this project include:

- NAPS School of IT as the internal academic client.
- Capstone project supervisor and academic assessors.
- BIT/MIT capstone students who will design, develop, test, and document the prototype.
- Students or staff who may use the system as learners during a demonstration.
- Administrators or academic staff who may manage awareness content and view quiz results.
- IT support staff who may provide general feedback on usability and cybersecurity relevance.

6. Constraints and Assumptions

The project scope should remain controlled so students can complete and demonstrate it within the capstone timeline.

- The system is an educational awareness prototype, not a live cybersecurity monitoring or phishing attack tool.
- Only sample training content, dummy users, and safe educational examples should be used.
- The system does not need to send real phishing emails or connect to real email accounts.
- No real passwords, sensitive personal data, or live institutional data should be collected.
- The quiz logic should be simple and rule-based; advanced AI or machine learning is not required.
- The project can be developed using common web development tools and a simple database.
- The final demonstration can use demo accounts, sample modules, quiz questions, and generated reports.

7. Expected Benefits

The expected benefits of the proposed project include:

- Provides students with a realistic and achievable cybersecurity-related capstone project.
- Demonstrates practical skills in web development, database design, content management, quiz logic, testing, and reporting.
- Improves basic cybersecurity awareness through simple training modules and quiz feedback.
- Creates a clear and demo-friendly system that can be presented effectively during the final assessment.
- Allows students to show both technical implementation and understanding of cybersecurity awareness principles.
- Can be extended later with additional topics, improved reports, or user progress tracking.

8. Available Required Resources

The following resources are sufficient for students to complete the project:

- Software: HTML, CSS, JavaScript, PHP/MySQL, Python Flask/Django, SQLite/MySQL, or another approved web development stack.
- Content: sample cybersecurity awareness topics, safe sample scenarios, and multiple-choice quiz questions.
- Dashboard tools: simple tables, charts, or Bootstrap-based admin pages.
- Hardware: standard student laptop or desktop computer.
- Data: dummy users, sample quiz attempts, and sample completion records.

- Documentation: user stories, database design, screenshots, testing checklist, user guide, and final project report.

9. Any Additional Information

This project is intentionally designed to be easy to implement and demonstrate. The minimum viable system should include training topic pages, multiple-choice quizzes, score calculation, feedback, admin question management, and a simple report or progress dashboard. Optional enhancements may include a printable certificate, module completion badges, improved charts, or a small password-strength awareness checker that does not store real passwords. Students should avoid any activity that involves real phishing, real credentials, or live institutional systems.

The End